

Discrete-Time Processing Overview

- Definitions
- Basic mathematical signals
- Signal types
- Discrete-time transforms: DFS, DTFT, DFT & FFT, z
- Energy spectrum
- DFT/FFT Interpretations
- Zero padding

Introduction

- Review of key discrete-time concepts
- Few examples
- Assume you are familiar with most of these ideas

Signals

- **Signals** are “physical quantities that change as a function of time, space, or some other independent variable.” (textbook)
- I will use **signal** to describe both physical signals and mathematical models of signals (e.g. $x(n) = \cos(\omega n)$).
- We will discuss both real-valued and complex-valued signals
 - Most signals are real
 - Some are not: e.g. $x(n) = e^{j\omega n}$
- The independent variable n is always real
- The signals may be vectors (arrays): $\mathbf{x}(n)$
 - Will denote in lower-case bold, but not upright
 - Matrices will be in upper-case bold
 - May be difficult to see on-screen
- Signals may be continuous-time, $x(t)$, or discrete-time, $x(n)$
- This class will only cover discrete-time signals

Digital Signals & Quantization

- The term **digital** is usually reserved for signals that are both discrete in amplitude (quantized) and time
- We will not address quantization
- Hence **discrete-time** is a more accurate description
- We assume that the amplitude of the (possibly noisy) signal $x(n)$ is known exactly
- More precisely, we assume the quantization error is negligible
- Under some general conditions quantization can be treated as a type of random noise and can be addressed using the techniques we will discuss

Signal Models

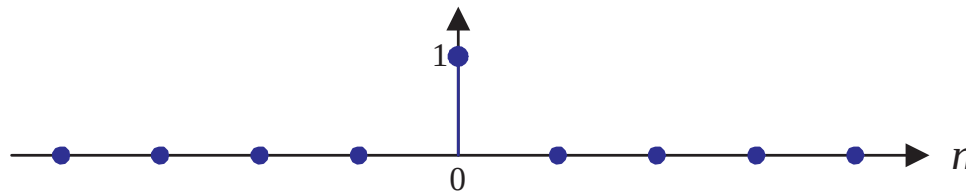
- Signals recorded from natural phenomena have no precise mathematical description
- Mathematical analysis requires a precise mathematical description
- To reconcile, we use models
 - The models are not exact
 - But are mathematically useful
 - Enable us to solve approximate problems exactly

Signal Classes

There are two broad classes of signals

- **Deterministic:** Any signal that can be described by a mathematical relationship is called
 - For example: $x(n) = \sin(\omega n)$
- **Random:** signals that cannot be described accurately by a mathematical relationship
 - For example: $x(n)$ independently, identically distributed (i.i.d.), $x(n) \sim N(0, 1)$
- We will focus on deterministic signals and systems first
- These will be useful building blocks for describing the properties of random signals

Discrete-Time Unit Impulse

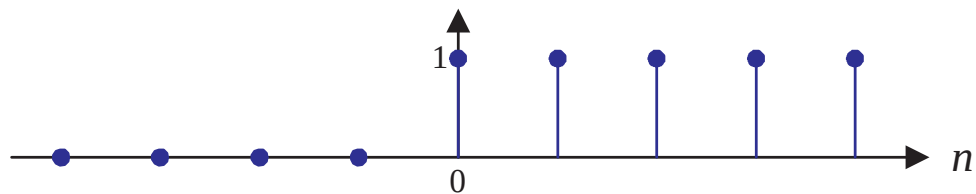


The discrete-time **unit impulse** is defined as

$$\delta(n) = \begin{cases} 0, & n \neq 0 \\ 1, & n = 0 \end{cases}$$

- Sometimes called the **unit sample**
- Also called the **Kronecker delta**
- Note that $\delta(n)$ is an even function so $\delta(n) = \delta(-n)$

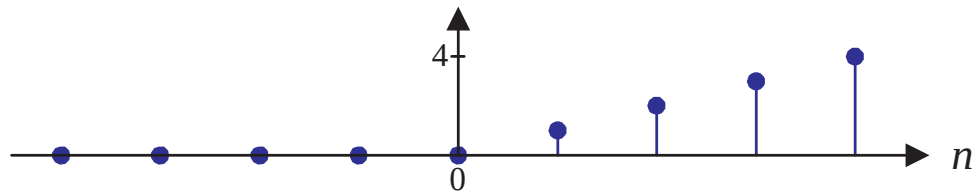
Discrete-Time Unit Step



The discrete-time **unit step** is defined as

$$u(n) = \begin{cases} 0, & n < 0 \\ 1, & n \geq 0 \end{cases}$$

Discrete-Time Unit Ramp



The discrete-time **unit ramp** is defined as

$$r(n) = n u(n) = \begin{cases} 0, & n < 0 \\ n, & n \geq 0 \end{cases}$$

Discrete-Time Basis Functions

- There is a close relationship between $\delta(n)$, $u(n)$ and $r(n)$

$$\delta(n) = u(n) - u(n-1) \qquad u(n) = r(n+1) - r(n)$$

$$u(n) = \sum_{k=-\infty}^n \delta(k) \qquad r(n) = \sum_{k=-\infty}^{n-1} u(k)$$

$$u(n) = \sum_{k=0}^{\infty} \delta(n-k) \qquad r(n) = \sum_{k=0}^{\infty} u(n-1-k)$$

- The unit impulse can be used to sample a discrete-time signal $x(n)$:

$$x(0) = \sum_{k=-\infty}^{\infty} x(k)\delta(k) \qquad x(n) = \sum_{k=-\infty}^{\infty} x(k)\delta(n-k)$$

- This ability to use the unit impulse to extract a single value of $x(n)$ is the basis of discrete-time convolution

Exponential Sequence

The discrete-time **exponential sequence** is defined as

$$x(n) = a^n$$

If a is complex,

$$a = re^{j\omega}$$

then $x(n)$ is a complex-valued signal and may be written as

$$x(n) = r^n e^{j\omega n} = x_R(n) + jx_I(n)$$

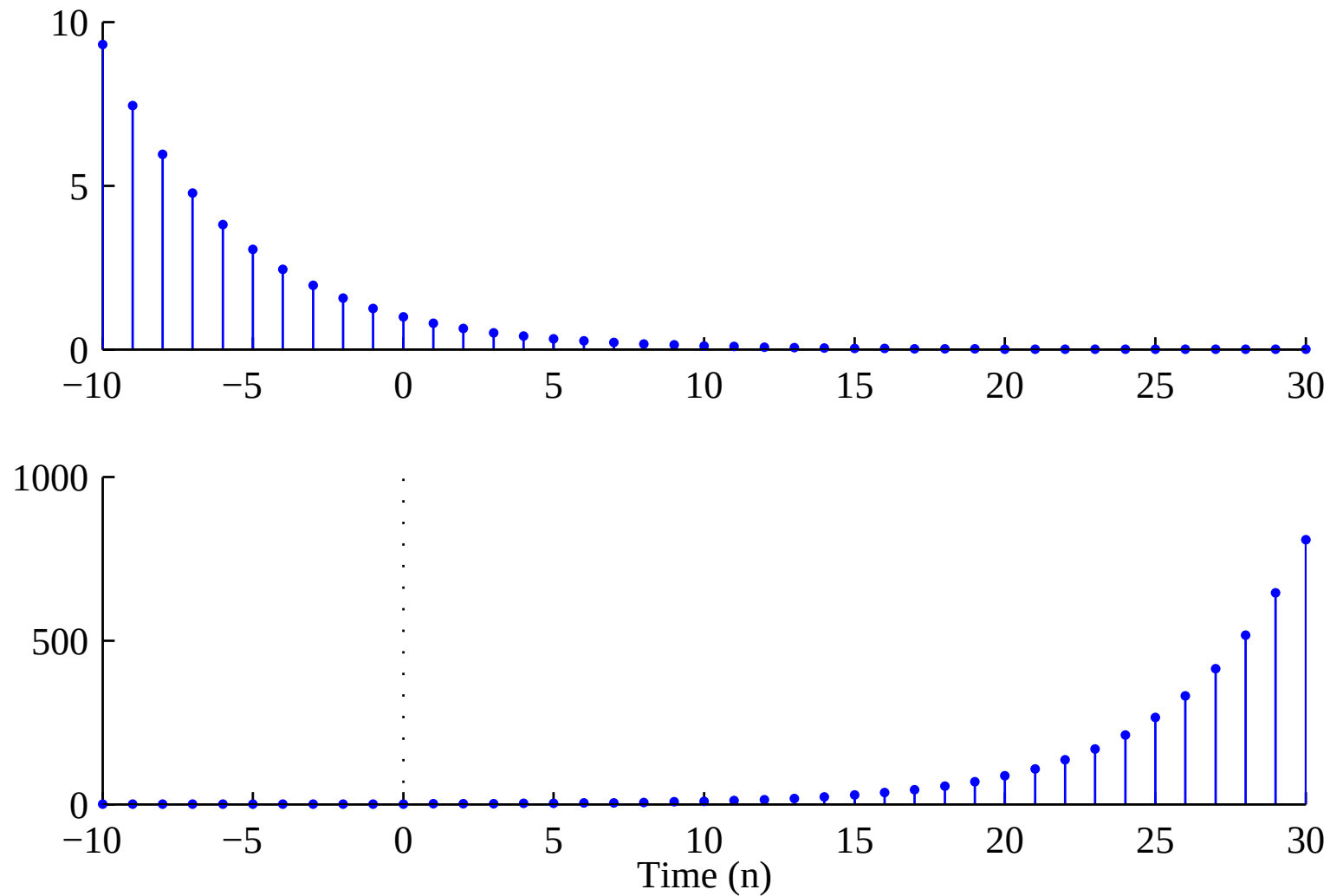
Recall Euler's identity

$$e^{j\omega n} = \cos(\omega n) + j \sin(\omega n)$$

Thus

$$x_R(n) = r^n \cos(\omega n) \qquad x_I(n) = r^n \sin(\omega n)$$

Example 1: 0.8^n and 0.8^{-n}



Example 1: MATLAB Code

```
function [] = ExponentialReal();

n = -10:30;          % Time index
a = 0.8;

figure;
FigureSet;
drawnow;
subplot(2,1,1);
    y = a.^(n); % Growing exponential
    h = stem(n,y);
    set(h(1),'Marker','.');
    set(gca,'Box','Off');
subplot(2,1,2);
    y = a.^(-n); % Decaying exponential
    h = stem(n,y);
    set(h(1),'Marker','.');
    set(gca,'Box','Off');
    xlabel('Time (n)');

AxisSet(8);
AxisLines;
set(gcf,'PaperPosition',[0 0 3 1.854]*1.5);
print -depsc ExponentialReal;

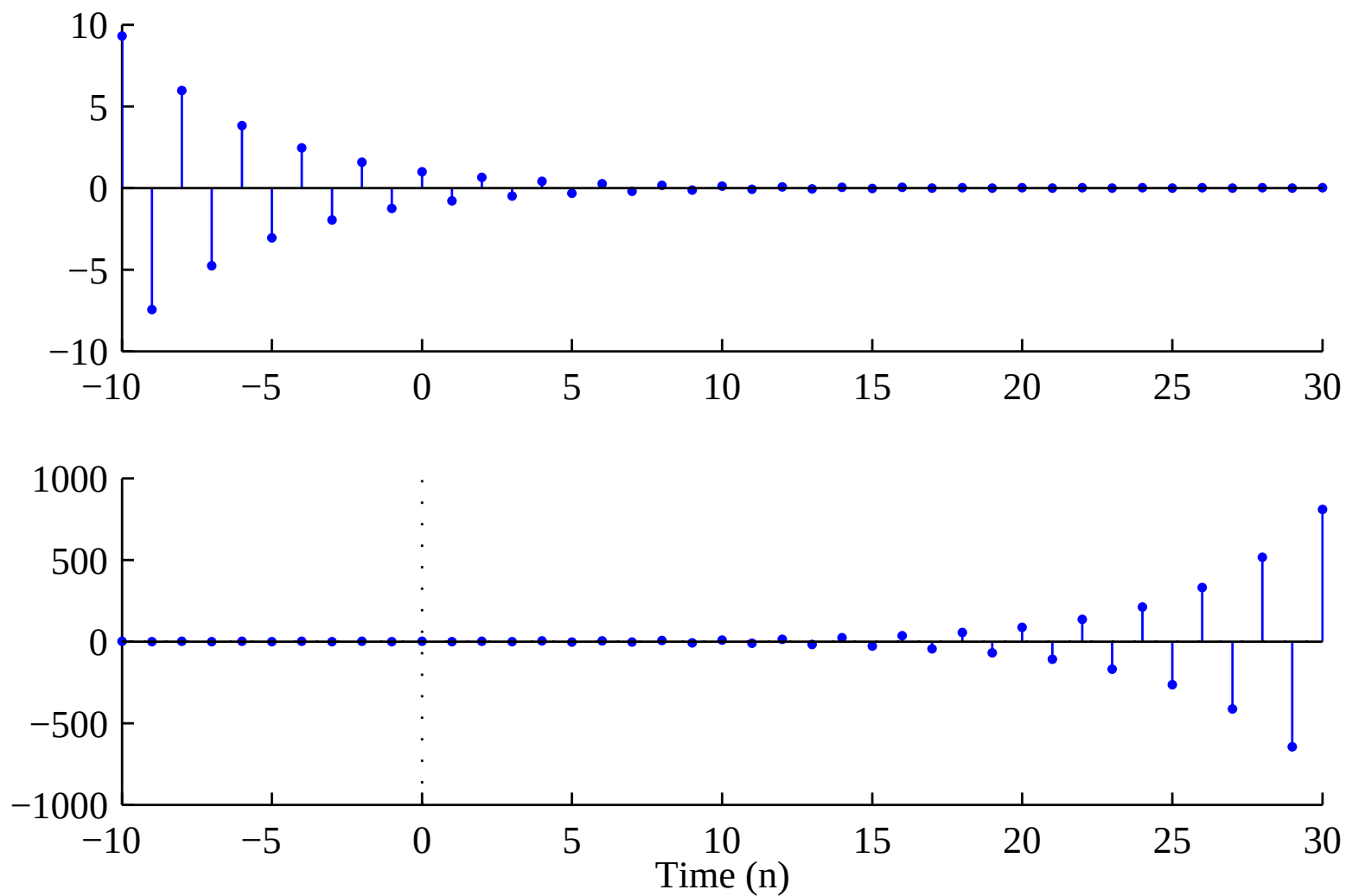
figure;
FigureSet;
```

```

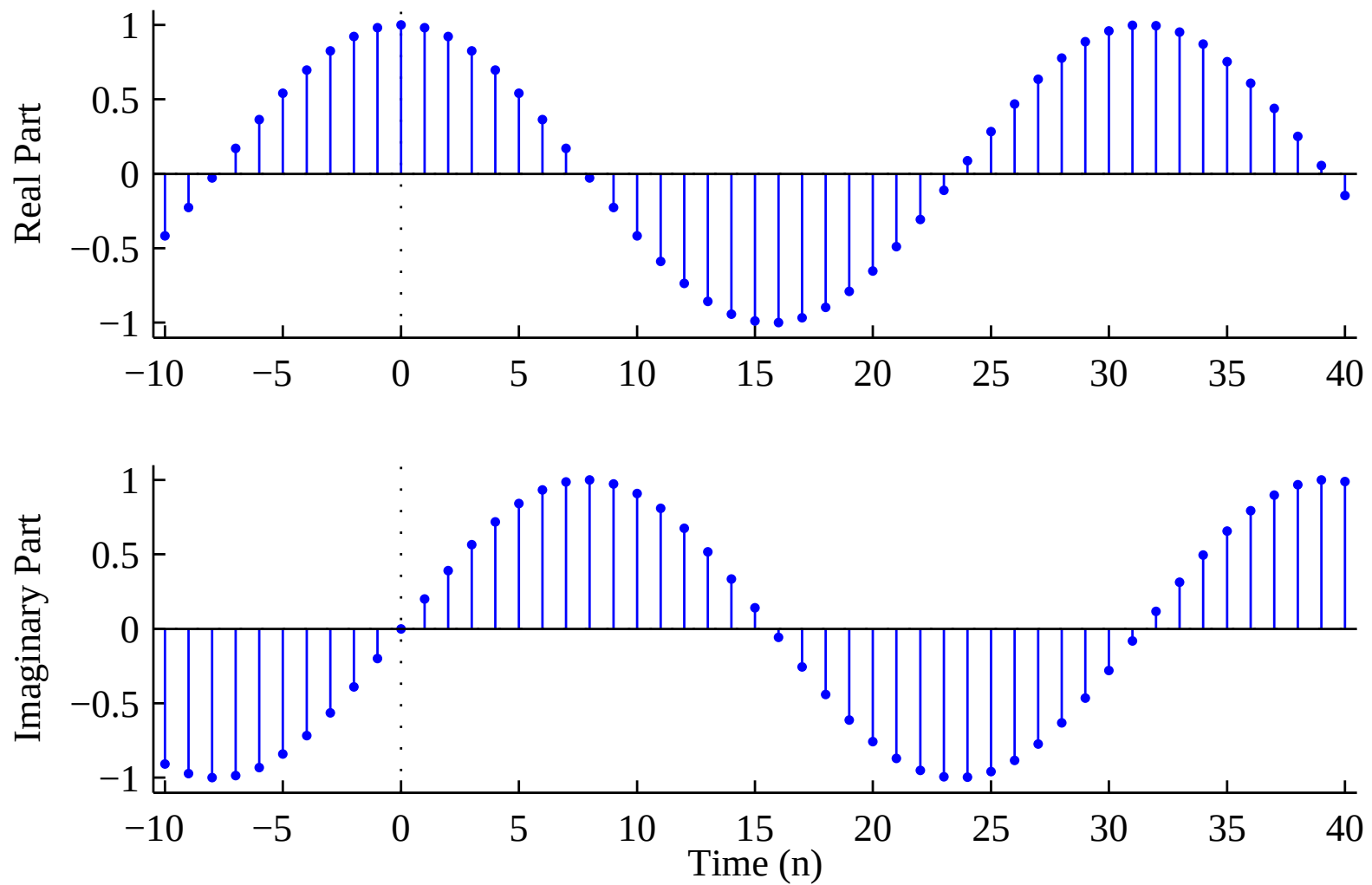
drawnow;
subplot(2,1,1);
    y = (-a).^(n); % Growing exponential
    h = stem(n,y);
    set(h(1),'Marker','.');
    set(gca,'Box','Off');
subplot(2,1,2);
    y = (-a).^(-n); % Decaying exponential
    h = stem(n,y);
    set(h(1),'Marker','.');
    set(gca,'Box','Off');
    xlabel('Time (n)');
AxisSet(8);
AxisLines;
set(gcf,'PaperPosition',[0 0 3 1.854]*1.5);
print -depsc ExponentialRealOscillating;

```

Example 2: $(-0.8)^n$ and $(-0.8)^{-n}$

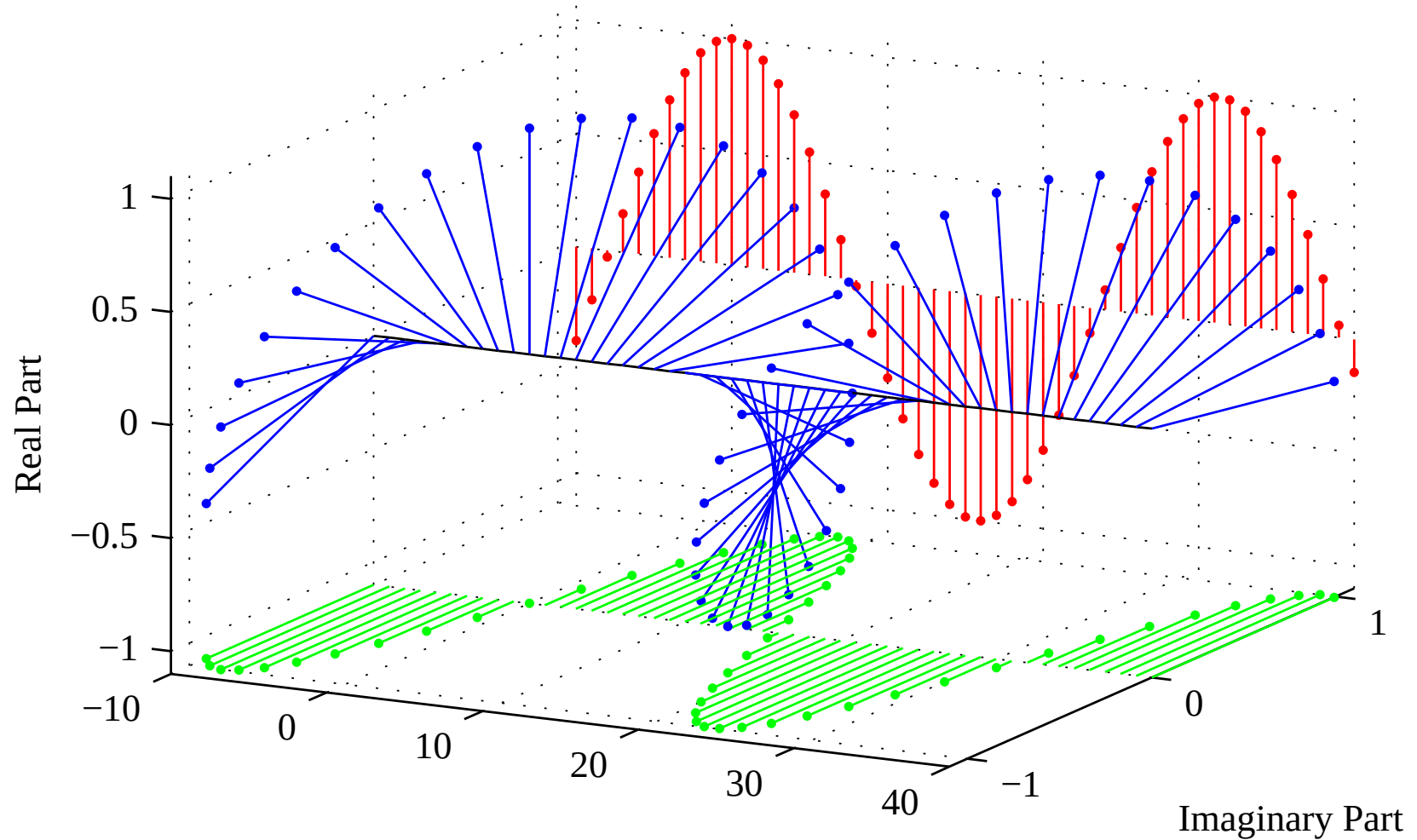


Example 3: $e^{j0.2n}$



Example 4: $e^{j0.2n}$

Complex:Blue Real:Red Imaginary:Green



Example 4: MATLAB Code

```
function [] = ExponentialImaginary();
close all;

figure;
FigureSet(1);

n = -10:40;           % Time index
N = length(n);
w = 0.2;
y = exp(j*w*n);
subplot(2,1,1);
    h = stem(n,real(y));
    set(h(1),'Marker','.');
    box off;
    xlim([n(1)-0.5 n(end)+0.5]);
    ylim([-1.1 1.1]);
    AxisLines;
    ylabel('Real Part');
subplot(2,1,2);
    h = stem(n,imag(y));
    set(h(1),'Marker','.');
    box off;
    xlim([n(1)-0.5 n(end)+0.5]);
    ylim([-1.1 1.1]);
    AxisLines;
    xlabel('Time (n)');
    ylabel('Imaginary Part');
```

```

AxisSet(8);
set(gcf,'PaperPosition',[0 0 3 1.854]*1.5);
print -depsc ExponentialImaginary;

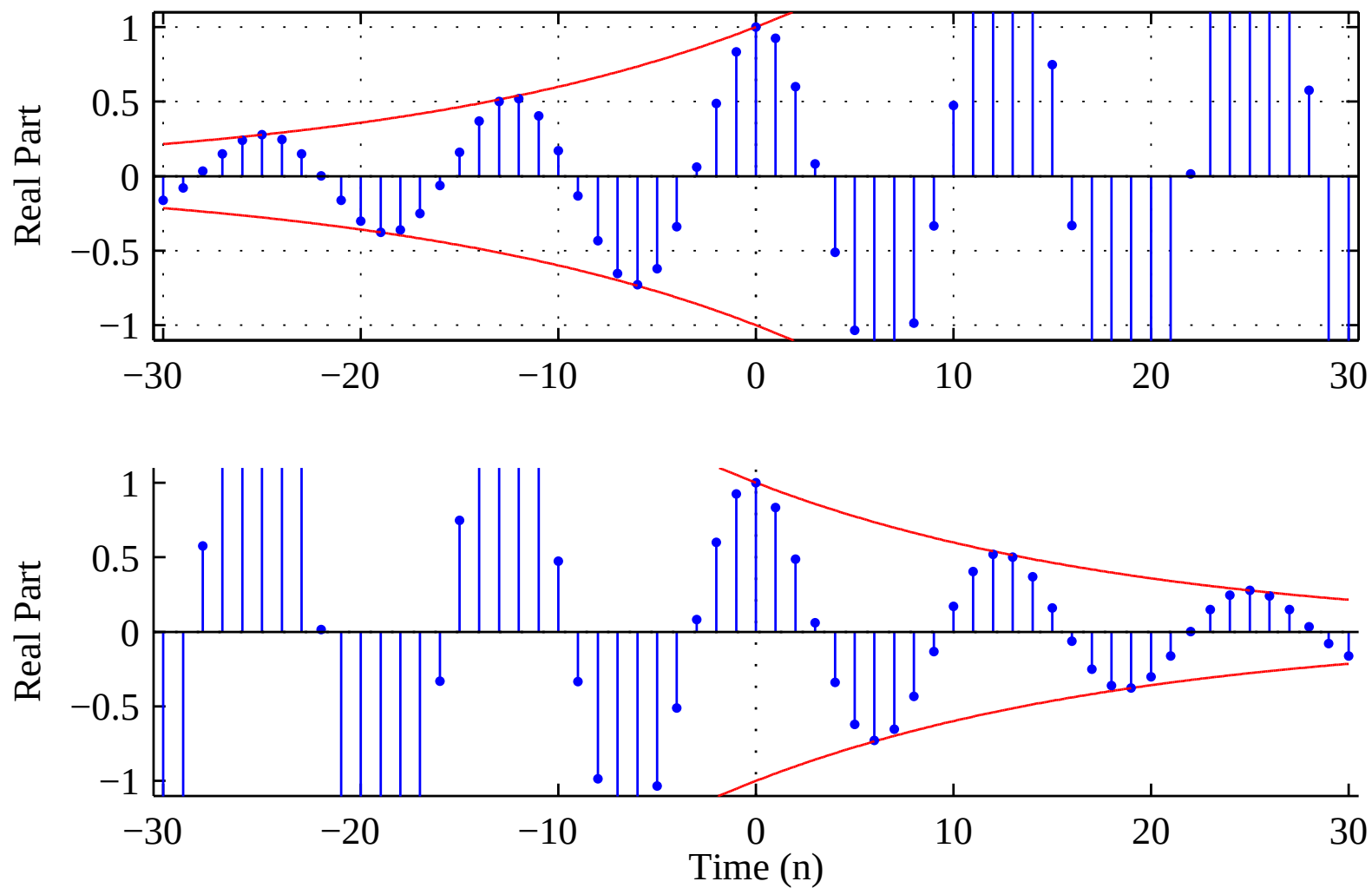
figure;
FigureSet(2);

h = plot3(n,zeros(1,N),zeros(1,N),'k');
hold on;
    h = plot3(ones(2,1)*n,[zeros(1,N);imag(y)],[zeros(1,N);real(y)],'b');
    h = plot3(n,imag(y),real(y),'b. ');
    h = plot3(ones(2,1)*n,1.1*ones(2,N),[zeros(1,N);real(y)],'r');
    h = plot3(n,1.1*ones(1,N),real(y),'r. ');
    h = plot3(ones(2,1)*n,[zeros(1,N);imag(y)],-1.1*ones(2,N),'g');
    h = plot3(n,imag(y),-1.1*ones(size(n)),'g. ');
hold off;
grid on;
ylabel('Imaginary Part');
xlabel('Real Part');
title('Complex:Blue   Real:Red   Imaginary:Green');
axis([min(n) max(n) -1.1 1.1 -1.1 1.1]);
view(27.5,22);

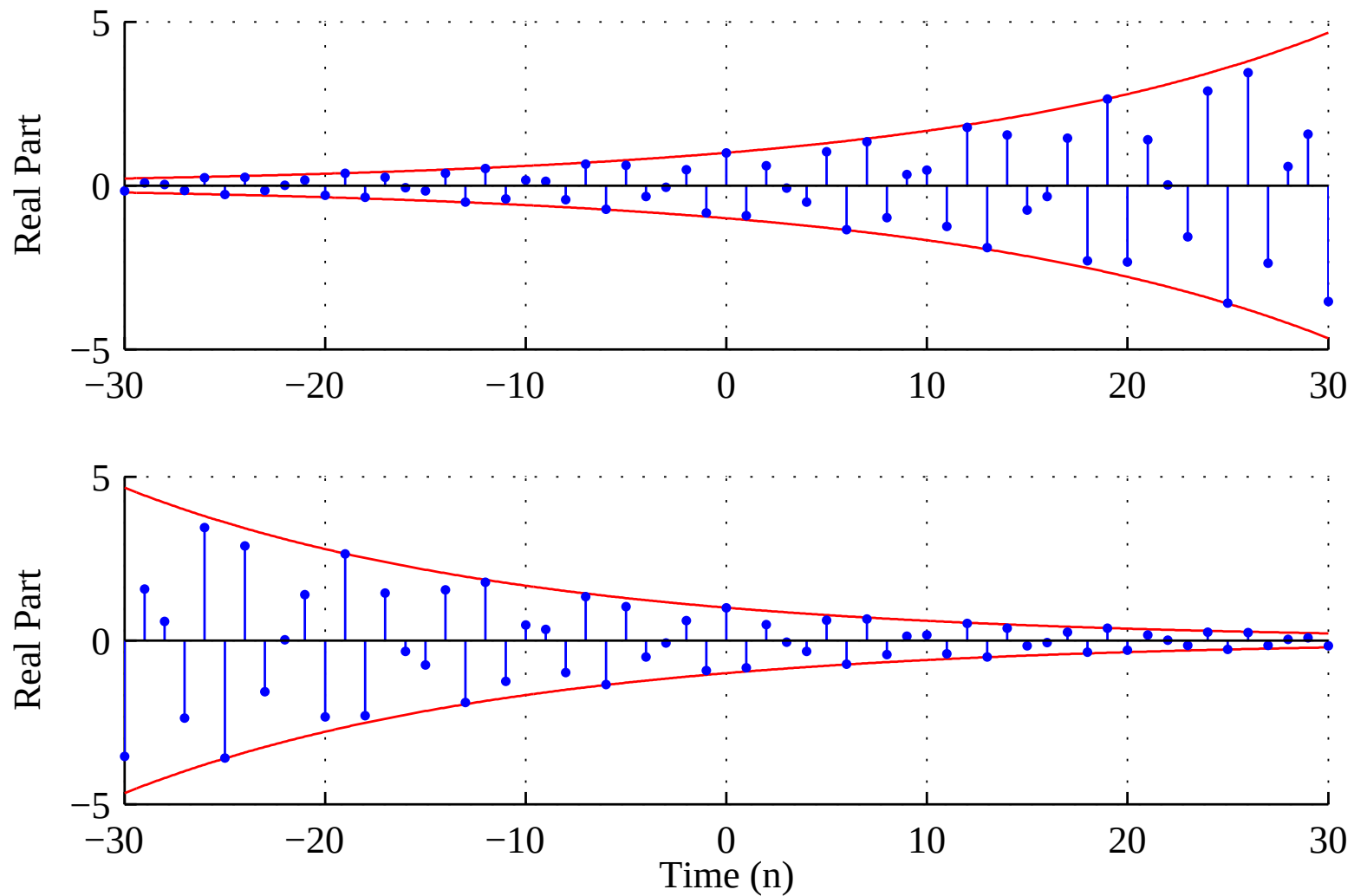
AxisSet(8);
set(gcf,'PaperPosition',[0 0 3 1.854]*1.5);
print -depsc ExponentialImaginary3D;

```

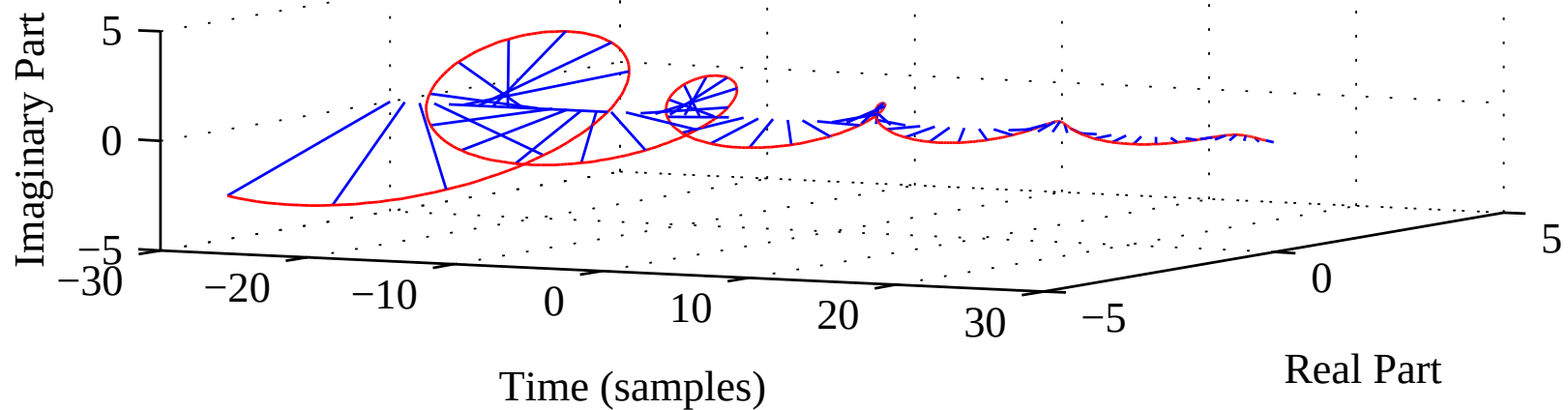
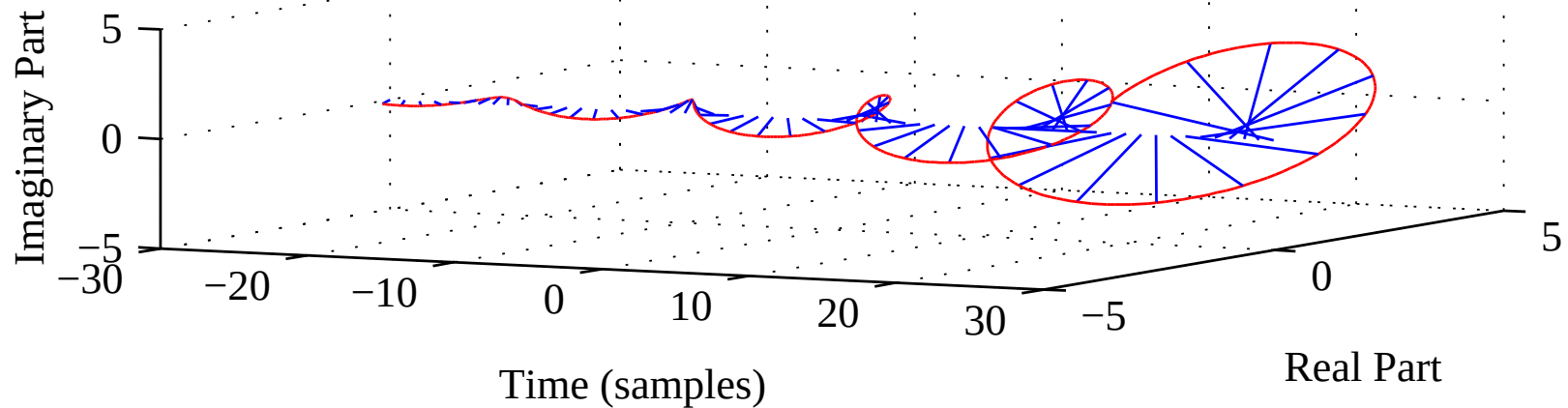
Example 5: $0.95^{\pm n} e^{j0.5n}$



Example 6: $(-0.95)^{\pm n} e^{j0.5n}$



Example 7: $0.95^{\pm n} e^{j0.5n}$



Example 7: MATLAB Code

```
%function [] = ExponentialComplex();
close all;

figure;
FigureSet(1);

a = 0.95;
w = 0.5;
n = -30:30;          % Time index
t = min(n):0.1:max(n);

subplot(2,1,1);
    x = real(a.^(-n).*exp(j*w*n));    % Growing exponential
    ev = real(a.^(-t));                % Envelope
    h = plot(t,-ev,'r',t,ev,'r');
    hold on;
        h = stem(n,x);
        set(h(1),'Marker','.');
    hold off;
    axis([n(1)-0.5 n(end)+0.5 -1.1 1.1]);
    AxisLines;
    grid on;
    ylabel('Real Part');
subplot(2,1,2);
    x = real(a.^(n).*exp(j*w*n));    % Growing exponential
    ev = real(a.^(t));                % Envelope
    h = plot(t,-ev,'r',t,ev,'r');
```

```

    hold on;
        h = stem(n,x);
        set(h(1), 'Marker', '.');
        hold off;
    box off;
    axis([n(1)-0.5 n(end)+0.5 -1.1 1.1]);
    AxisLines;
    xlabel('Time (n)');
    ylabel('Real Part');

AxisSet(8);
set(gcf, 'PaperPosition', [0 0 3 1.854]*1.5);

print -depsc ExponentialComplexRealPart;

figure;
FigureSet;
subplot(2,1,1);
    x = real((-a).^(-n).*exp(j*w*n)); % Growing exponential
    ev = real(a.^(-t)); % Envelope
    h = plot(t,-ev, 'r', t, ev, 'r');
    hold on;
        h = stem(n,x);
        set(h(1), 'Marker', '.');
        hold off;
    box off;
    grid on;
    ylabel('Real Part');
subplot(2,1,2);
    x = real((-a).^(n).*exp(j*w*n)); % Growing exponential
    ev = real(a.^(t)); % Envelope

```



```

h = plot(t,-ev,'r',t,ev,'r');
hold on;
    h = stem(n,x);
    set(h(1),'Marker','.');
    hold off;
box off;
grid on;
xlabel('Time (n)');
ylabel('Real Part');

AxisSet(8);
set(gcf,'PaperPosition',[0 0 3 1.854]*1.5);

print -depsc ExponentialComplexRealPartOscillating;

figure;
FigureSet;

subplot(2,1,1);
    xd = a.^(-n).*exp(j*w*n);           % Growing exponential, discrete-time
    xc = a.^(-t).*exp(j*w*t);           % Growing exponential, continuous-time
    h = plot3(t,real(xc),imag(xc),'r');
    hold on;
        for c1 = 1:length(n),
            h = plot3([n(c1) n(c1)], [0 real(xd(c1))], [0 imag(xd(c1))], 'b');
        end;
    hold off;
box off;
grid on;
xlabel('Time (samples)');
ylabel('Real Part');

```

```

    xlabel('Imaginary Part');
    view(27.5,22);
subplot(2,1,2);
    xd = a.^(n).*exp(j*w*n);           % Growing exponential, discrete-time
    xc = a.^(t).*exp(j*w*t);           % Growing exponential, continuous-time
    h = plot3(t,real(xc),imag(xc),'r');
    hold on;
        for c1 = 1:length(n),
            h = plot3([n(c1) n(c1)],[0 real(xd(c1))],[0 imag(xd(c1))],'b');
        end;
    hold off;
    box off;
    grid on;
    xlabel('Time (samples)');
    ylabel('Real Part');
    xlabel('Imaginary Part');
    view(27.5,22);

AxisSet(8);
set(gcf,'PaperPosition',[0 0 3 1.854]*1.5);

print -depsc ExponentialComplex3D;

```

Continuous-Time Unit Impulse

$$\delta(\omega) \equiv \begin{cases} 0 & \omega \neq 0 \\ \infty & \omega = 0 \end{cases} \quad \int_{-e}^e \delta(\omega) d\omega = 1 \text{ for any } e > 0$$

- Called **continuous-time** unit impulse
- Also known as the **Dirac delta** function
- Is zero everywhere except zero
- The impulse integral serves as a measure of the impulse amplitude
- May think of as the derivative of a continuous-time unit step
- Drawn as an arrow with unit height
- $5\delta(\omega)$ would be drawn as an arrow with height of 5

Continuous-Time Unit Impulse Properties

$$\delta(\omega)x(\omega) = \delta(\omega)x(0)$$

$$\delta(a\omega) = \frac{1}{|a|}\delta(\omega)$$

$$\delta(-\omega) = \delta(\omega)$$

$$\delta(\omega) = \frac{du(\omega)}{d\omega}$$

$$u(\omega) = \int_{-\infty}^{\omega} \delta(u) du$$

Signal Categories: Energy and Power

- **Energy Signal:** signals such that the energy,

$$E_x = \sum_{n=-\infty}^{\infty} |x(n)|^2$$

is finite: $0 < E_x < \infty$

– Note my definition differs from text slightly to include $x(n) = 0$

- **Power Signals:** signals such that the average power,

$$P_x = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x(n)|^2$$

is finite and non-zero: $0 < P_x < \infty$

- Signals that are finite in duration and amplitude are energy signals
- Energy signals have zero power
- Power signals have infinite energy

Signal Categories: Periodic and Aperiodic

- **Periodic:** signals such that

$$x(n) = x(n + N)$$

for some integer N .

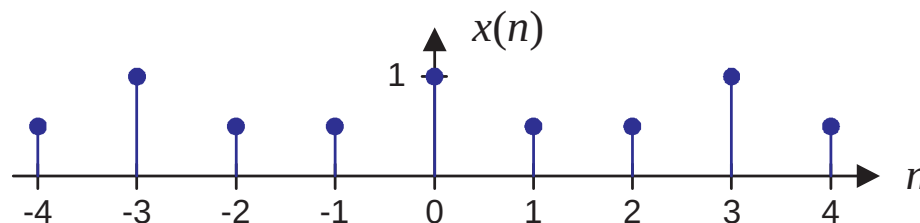
- **Aperiodic:** signals that are not periodic
- Note the complex exponential (and sinusoidal signals in general) is periodic iff (if and only if)

$$\frac{\omega}{2\pi} = \frac{k}{N}$$

In other words, $\frac{\omega}{2\pi}$ is a rational number

- This is not true for the continuous-time counterpart in which N is allowed to take on fractional values
- **Almost Periodic:** signals such that $x(n) = x(n + M)$ for some irrational number M

Definition of Discrete-Time Periodic Signals



- A signal $x(n)$ is **periodic** if there exists a non-zero, positive value of N such that

$$x(n + N) = x(n) \text{ for all } n$$

- The **fundamental period** N_0 is the smallest positive value of N for which the above holds
- The **fundamental frequency** is
 - $f_0 = \frac{1}{N_0} = \frac{\omega_0}{2\pi}$ cycles/sample
 - $\omega_0 = \frac{2\pi}{N_0} = 2\pi f_0$ radians/sample
- Will frequently drop the 0 subscript to simplify notation

Signal Categories: Duration and Causality

- **Finite duration:** signal has this property if there exists a finite N_1 and N_2 such that $x(n) = 0$ for $n < N_1$ and $n > N_2$
- **Infinite duration:** signal that does not have a finite duration.
- **Causal:** a signal is said to be causal if $x(n) = 0$ for $n < 0$
- **Anti-Causal:** a signal is said to be anti-causal if $x(n) = 0$ for $n > 0$
- **Non-Causal:** a signal is said to be non-causal if $x(n) \neq 0$ for some $n < 0$. In other words, the signal is not causal.

Signal Categories Even & Odd Symmetry

$$x_e(n) = \frac{1}{2} (x(n) + x(-n))$$

$$x_o(n) = \frac{1}{2} (x(n) - x(-n))$$

$$x(n) = x_e(n) + x_o(n)$$

- **Even:** a signal is **even** if and only if $x(n) = x(-n)$
- **Odd:** a signal is **odd** if and only if $x(n) = -x(-n)$
- $\cos(k\omega_0 n)$ is an even signal
- $\sin(k\omega_0 n)$ is an odd signal
- Any signal can be written as the sum of an odd signal and an even signal

Random Signals

- **Random Signals:** Signals that “evolve” in time in an unpredictable manner
- Cannot be described by explicit mathematical relationships
- Many examples
 - Thermal noise
 - Stock market
 - Turbulence
- Includes signals in which the physics are too complex to model exactly
- In order to work with random signals, we must assume something does not change
- In this class, we will assume the **average properties** do not change (the signals are locally stationary)
- We will analyze these signals using statistical methods

Discrete-Time Exponential Harmonics

- The set of all discrete-time complex exponential signals that are periodic with period N can be expressed as

$$\phi_k(n) = e^{jk\omega n} = e^{jk\frac{2\pi}{N}n}, \quad k = 0, \pm 1, \pm 2, \dots$$

- $e^{j\omega n}$ is called the **fundamental component**
- $e^{j2\omega n}$ is usually called the 2nd **harmonic component**
- $e^{jk\omega n}$ is usually called the k th **harmonic component**

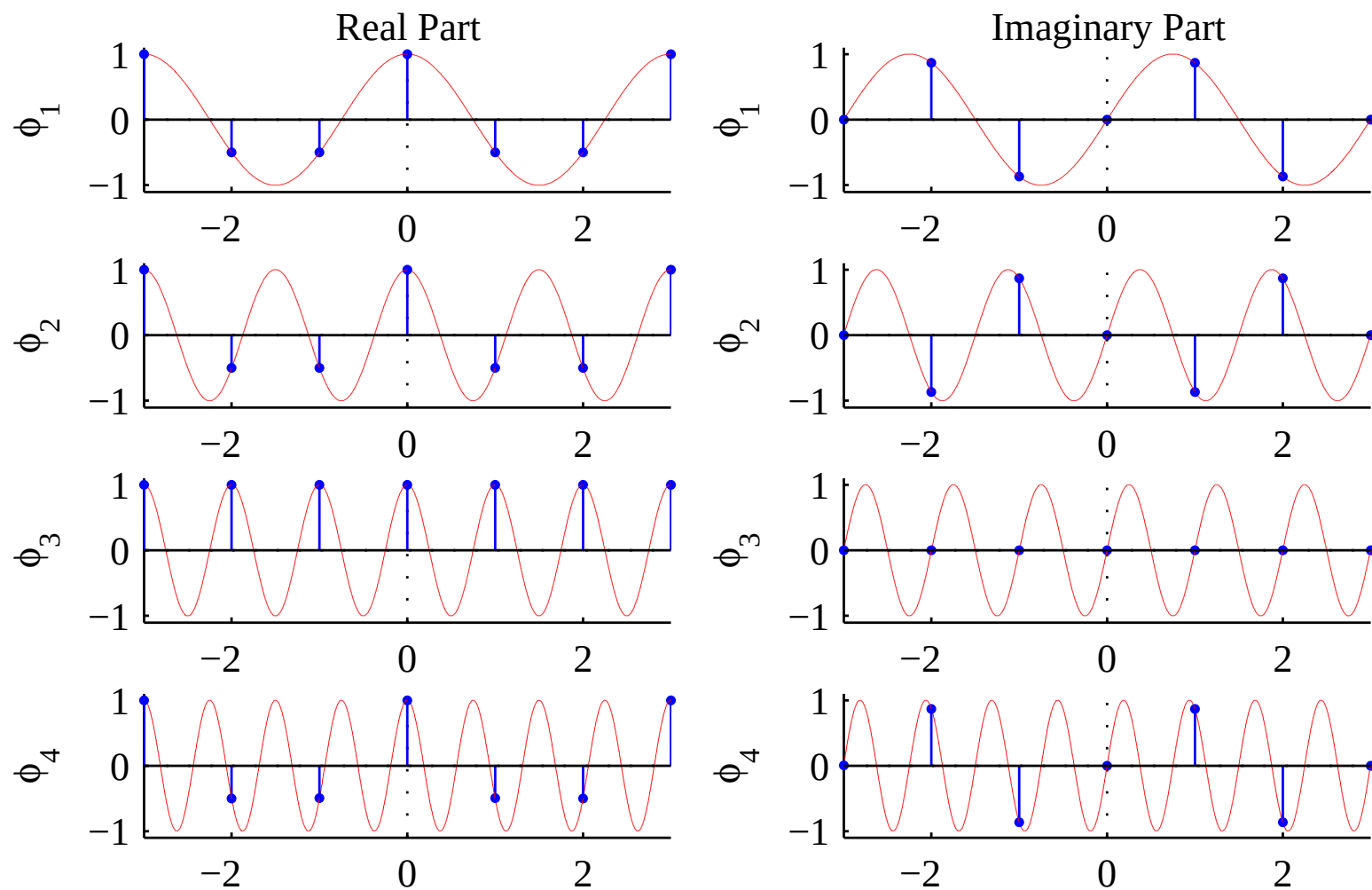
Discrete-Time Exponential Harmonics Redundancy

Unlike continuous-time exponentials, there are only N distinct harmonics

$$\begin{aligned}\phi_k(n) &= e^{jk \frac{2\pi}{N} n} \\ \phi_{k+rN}(n) &= e^{j(k+rN) \frac{2\pi}{N} n} \\ &= e^{jk \frac{2\pi}{N} n + jr2\pi n} \\ &= e^{jk \frac{2\pi}{N} n} e^{jr2\pi n} \\ &= e^{jk \frac{2\pi}{N} n} \\ &= \phi_k(n)\end{aligned}$$

Thus, we cannot distinguish between the k th harmonic and the $(k + rN)$ th harmonic. This is an important difference between discrete-time and continuous-time signals and is the basis of **aliasing**.

Example 8: Discrete-Time Harmonics ($N = 3$)



Example 8: MATLAB Code

```
function [] = CTHarmonics();
close all;

figure;
FigureSet(1);

N = 3;
w = 2*pi/N;
fs = 500;           % Real-Time Sample rate (Hz)
t = -3:1/fs:3;      % Time index (s)
n = -3:1:3;         % Sample index

for cnt = 1:4,
    subplot(4,2,cnt*2-1);
    h = plot(t,real(exp(j*cnt*w*t)),'r');
    set(h,'LineWidth',0.1);
    hold on;
    h = stem(n,real(exp(j*cnt*w*n)),'.');
    hold off;
    box off;
    %grid on;
    xlim([min(t) max(t)]);
    ylim([-1.1 1.1]);
    AxisLines;
    ylabel(sprintf('\\phi_%d',cnt));
if cnt==1,
    title('Real Part');
```

```

        end;

subplot(4,2,cnt*2);
    h = plot(t,imag(exp(j*cnt*w*t)),'r');
    set(h,'LineWidth',0.1);
    hold on;
        h = stem(n,imag(exp(j*cnt*w*n)),'.');
        hold off;
    box off;
    %grid on;
    xlim([min(t) max(t)]);
    ylim([-1.1 1.1]);
    AxisLines;
    ylabel(sprintf('\\phi_%d',cnt));
if cnt==1,
    title('Imaginary Part');
end;
end;

```

```

AxisSet(8);
set(gcf,'PaperPosition',[0 0 3 1.854]*1.5);
print -depsc DTHarmonics;

```

Exponential Harmonics are Orthogonal

$$\sum_{n=\langle N \rangle} e^{jk\omega_o n} = \begin{cases} N & k = 0, \pm N, \pm 2N, \dots \\ 0 & \text{otherwise} \end{cases}$$

- This property is one of the key reasons that Fourier analysis is so powerful and useful

Discrete-Time Fourier Series

- A sum of harmonically related exponentials still has the fundamental frequency ω

$$x(n) = \sum_{k=\langle N \rangle} a_k e^{jk\omega n} = \sum_{k=\langle N \rangle} a_k e^{jk\frac{2\pi}{N}n}$$

- The book uses X_k instead of a_k and limits the range from $k = 0$ to $n = N - 1$
- Any periodic signal $x(n)$ with fundamental period N can be written in this form
- $x(n)$ is defined by N unique values during a single period
- The sum contains N distinct terms with N complex-valued coefficients
- Symmetry limits the degrees of freedom to N

Symmetry of the Coefficients

$$x(n) = \sum_{k=\langle N \rangle} a_k e^{jk\omega n} \quad a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x(n) e^{-jk\omega n}$$

- There are only N distinct discrete-time exponential harmonics
- Need N samples to calculate the DTFS coefficients a_k
- The coefficients (power spectrum) are also periodic: $a_k = a_{k+N}$
- Thus, a_k is also a discrete-time periodic signal (complex valued)
- If $x(n)$ is real, can also show that $a_k = a_{-k}^*$
- The DTFS always “converges” for all finite periodic signals $x(n)$
- Parseval’s theorem

$$P_x = \frac{1}{N} \sum_{n=\langle N \rangle} |x(n)|^2 = \sum_{k=\langle N \rangle} |a_k|^2$$

- The sequence $|a_k|^2$ called the **power spectrum**

Discrete-Time Fourier Transform

$$X(e^{j\omega}) \triangleq \mathcal{F}\{x(n)\} = \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n}$$

$$x(n) = \mathcal{F}^{-1}\{X(e^{j\omega})\} = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n} d\omega$$

- $\omega = 2\pi f$ (*radians per sample*), f has units of *cycles per sample*
- Called the **analysis** and **synthesis** equations
- Said to form a **Fourier transform pair**: $x(n) \xLeftrightarrow{\mathcal{FT}} X(e^{j\omega})$

Discrete-Time Fourier Transform Convergence

Any discrete-time signal that is absolutely summable or has finite energy

$$\sum_{n=-\infty}^{\infty} |x(n)| < \infty \quad \text{or} \quad \sum_{n=-\infty}^{\infty} |x(n)|^2 < \infty$$

can be described by the discrete-time Fourier transform.

- There are fewer convergence issues than in the continuous-time case
- It can be shown that absolutely summable signals have finite energy
- The converse is not true in general (e.g. $x(n) = 1/n$)

DTFT Properties

$$X(e^{j\omega}) \triangleq \mathcal{F}\{x(n)\} = \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n}$$

$$x(n) = \mathcal{F}^{-1}\{X(e^{j\omega})\} = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n} d\omega$$

- $X(e^{j\omega})$ is periodic with fundamental period 2π
- If $x(n)$ is real valued, then $X(e^{j\omega})$ has the following symmetry
 - $|X(e^{j\omega})|$ is even
 - $\angle X(e^{j\omega})$ is odd
 - $\text{Re}\{X(e^{j\omega})\}$ is even
 - $\text{Im}\{X(e^{j\omega})\}$ is odd

Parseval's Relation

Parseval's relation:

$$E_x = \sum_{n=-\infty}^{\infty} |x(n)|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$$

- $|X(e^{j\omega})|^2 \geq 0$ indicates the *energy density* at frequency ω
- This is similar to a probability density function (pdf) and has many equivalent properties
- Thus, it is called the **energy spectrum** of $x(n)$

DTFT Energy Spectrum

$$e^{j\omega_0 n} \xLeftrightarrow{\mathcal{FT}} \sum_{\ell=-\infty}^{\infty} 2\pi\delta(\omega - \omega_0 - 2\pi\ell)$$

where $\delta(\cdot)$ in this case is the dirac delta

- Almost periodic signals, such as $x(n) = A \cos(\omega_0 n + \theta)$ where $\frac{\omega}{2\pi}$ is not a rational function have line spectra at $\pm\omega_0 \pm 2\pi k$ for any integer k
- Thus, periodic and almost periodic signals may be treated the same in the frequency domain
- Periodic signals are power signals and have infinite energy
- A DC signal, $x(n) = c$ where c is a constant, are also power signals

Harmonic and Nonharmonic Spectra

- If $x(n)$ can be expressed as

$$x(n) = \sum_{\ell=1}^L a_{\ell} e^{j\omega_{\ell} n}$$

where the frequencies are nonharmonically related, then $x(n)$ has **discrete nonharmonic spectra**

- If harmonically related, then $x(n)$ is periodic and has **harmonic spectra**
- Signals with spectra that consist of both impulses and a continuous spectrum are called **mixed**

$$u(n) \xleftrightarrow{\mathcal{FT}} \frac{1}{1 - e^{-j\omega}} + \sum_{k=-\infty}^{\infty} \pi \delta(\omega - 2\pi k)$$

Discrete Fourier Transform (DFT) Importance

- The DFT is the workhorse of signal processing because it can be computed very efficiently with fast algorithms
- These fast algorithms are called **fast Fourier transform (FFT)** algorithms
- The DFT/FFT can be used for many mathematical operations
 - Convolution
 - DTFT
 - Signal filtering
 - Integration and differentiation
- FFT computation is on the order of $N \log_2 N$

Discrete Fourier Transform (DFT)

The discrete Fourier transform (DFT) of an N point sequence is defined as

$$X(k) = \sum_{n=0}^{N-1} x(n)e^{-j(2\pi/N)kn}$$
$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k)e^{j(2\pi/N)kn}$$

- Note that I use $X(k)$ for the DFT and the book uses $\tilde{X}(k)$
- Will be careful not to confuse with the DTFT $X(e^{j\omega})$
- Note the DFT only analyzes a finite segment of the signal $x(n)$
- Our interpretation of the DFT depends on how we interpret what happens outside of the observed range: $n < 0$ and $n \geq N$

DFT Interpretation: Finite Duration

If we assume that $x(n)$ really is of finite duration and is zero outside of the observed range, then the DFT is related to the DTFT as follows

$$\begin{aligned} X(e^{j\omega})|_{\omega=(2\pi/N)k} &= \sum_{n=0}^{N-1} x(n)e^{-j\omega n} \Big|_{\omega=(2\pi/N)k} \\ &= \sum_{n=0}^{N-1} x(n)e^{-j(2\pi/N)kn} \\ &= X(k) \end{aligned}$$

Thus the N -point DFT of a finite-duration signal with length N is equal to the Fourier transform of the signal at discrete frequencies

$$\omega_k = (2\pi/N)k \text{ for } 0 \leq k \leq N-1$$

DFT Interpretation: Windowing

If $x(n)$ is infinite duration and $w(n)$ is a finite duration window, say

$$w(n) = \begin{cases} 1 & 0 \leq n < N \\ 0 & \text{otherwise} \end{cases}$$

Then the product $x(n)w(n)$ is a discrete-time product and we know

$$x(n)w(n) \xLeftrightarrow{\mathcal{FT}} \frac{1}{2\pi} \int_{2\pi} X(e^{ju})W(e^{j(\omega-u)}) du$$

- In words, the sampled signal's spectrum is equal to the convolution of the signal's spectrum $X(e^{j\omega})$ and the window's spectrum $W(e^{j\omega})$
- This will be very important when we discuss spectral estimation
- Note that the convolution is circular

DFT Interpretation: Periodic Signals

If $x(n)$ is periodic with fundamental period N , then the DFT of one fundamental period of $x(n)$ is related to the Fourier series coefficients as follows

$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x(n) e^{-jk\omega n} = \frac{1}{N} X(k)$$

Thus, the DFT of one period of a periodic signal is proportional to the Fourier series coefficients of the signal.

Zero Padding

Let $x(n) = y(n)w(n)$. Then

$$X(e^{j\omega})|_{\omega=(2\pi/N)k} = \sum_{n=0}^{N-1} y(n)w(n)e^{-j(2\pi/N)kn} = X(k)$$

- The FFT has two apparent disadvantages
 - It requires that N be an integer power of 2: $N = 2^\ell$
 - It only generates estimates at N frequencies equally spaced between 0 and 2π rad/sample
- Both of these problems can be circumvented by **zero-padding**
- We can choose N to be larger than the length of our window $w(n)$

Zero Padding Derived

Suppose we add $M - N$ zeros to the finite-length signal $x(n)$ such that M is an integer power of 2 and $M \geq N$. Then the padded signal, $\acute{x}(n)$, has a length M . The frequency sampling density then improves to $\frac{2\pi}{M}$ rad/sample, rather than $\frac{2\pi}{N}$ rad/sample.

$$\text{DFT} \{x(n)\} = \sum_{n=0}^{N-1} x(n) e^{-j(k \frac{2\pi}{N})n}$$

$$\text{DFT} \{\acute{x}(n)\} = \sum_{n=0}^{M-1} \acute{x}(n) e^{-j(k \frac{2\pi}{M})n}$$

$$= \sum_{n=0}^{N-1} x(n) e^{-j(k \frac{2\pi}{M})n}$$

$$X(e^{j\omega}) \Big|_{\omega=k \frac{2\pi}{M}} = \text{DFT} \{\acute{x}(n)\}$$

Example 10: FFT Estimate of DTFT

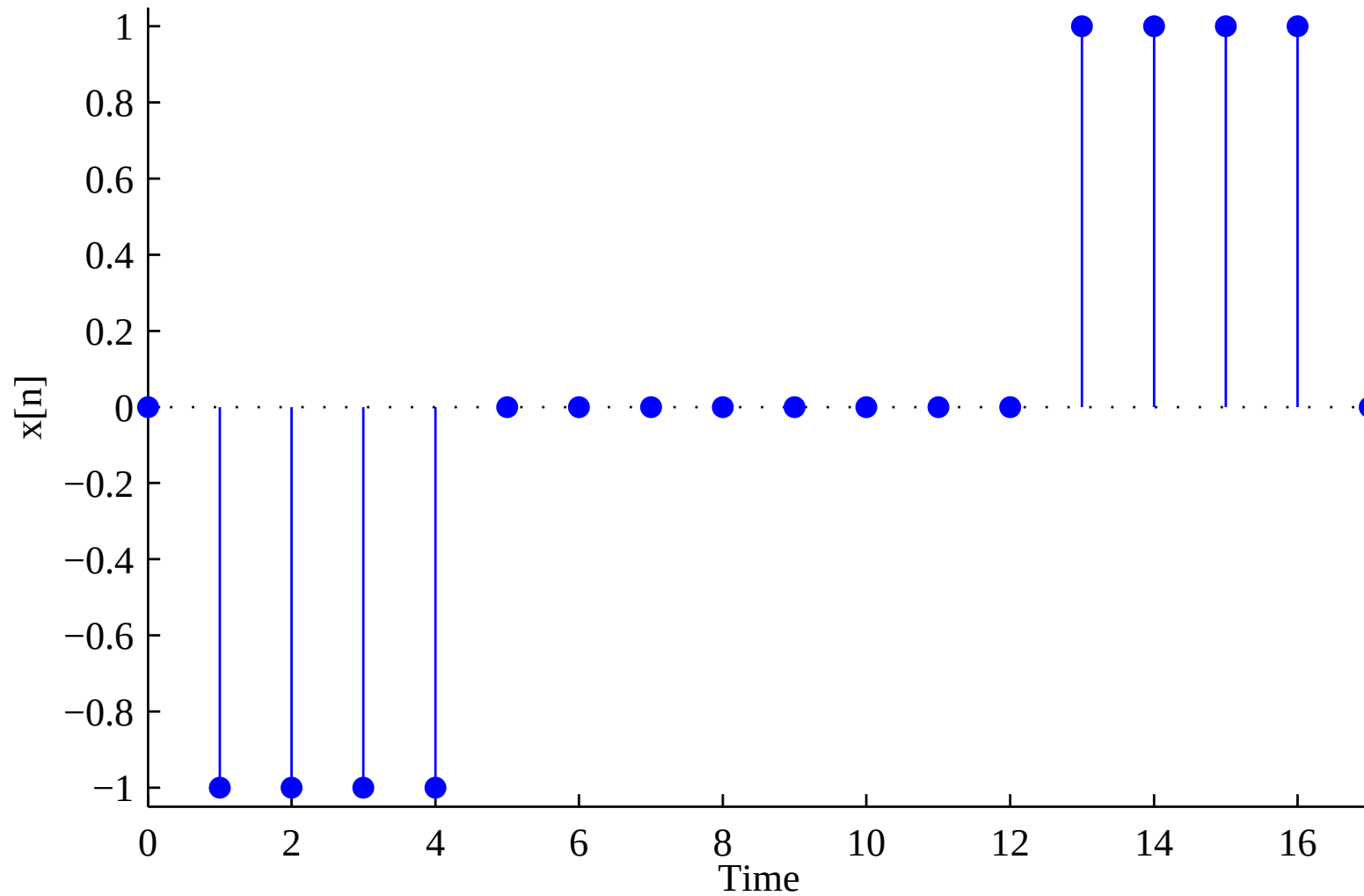
Use the FFT to calculate the DTFT of

$$x(n) = \begin{cases} -1 & 1 \leq n \leq 4 \\ +1 & 13 \leq n \leq 16 \\ 0 & \text{Otherwise} \end{cases}$$

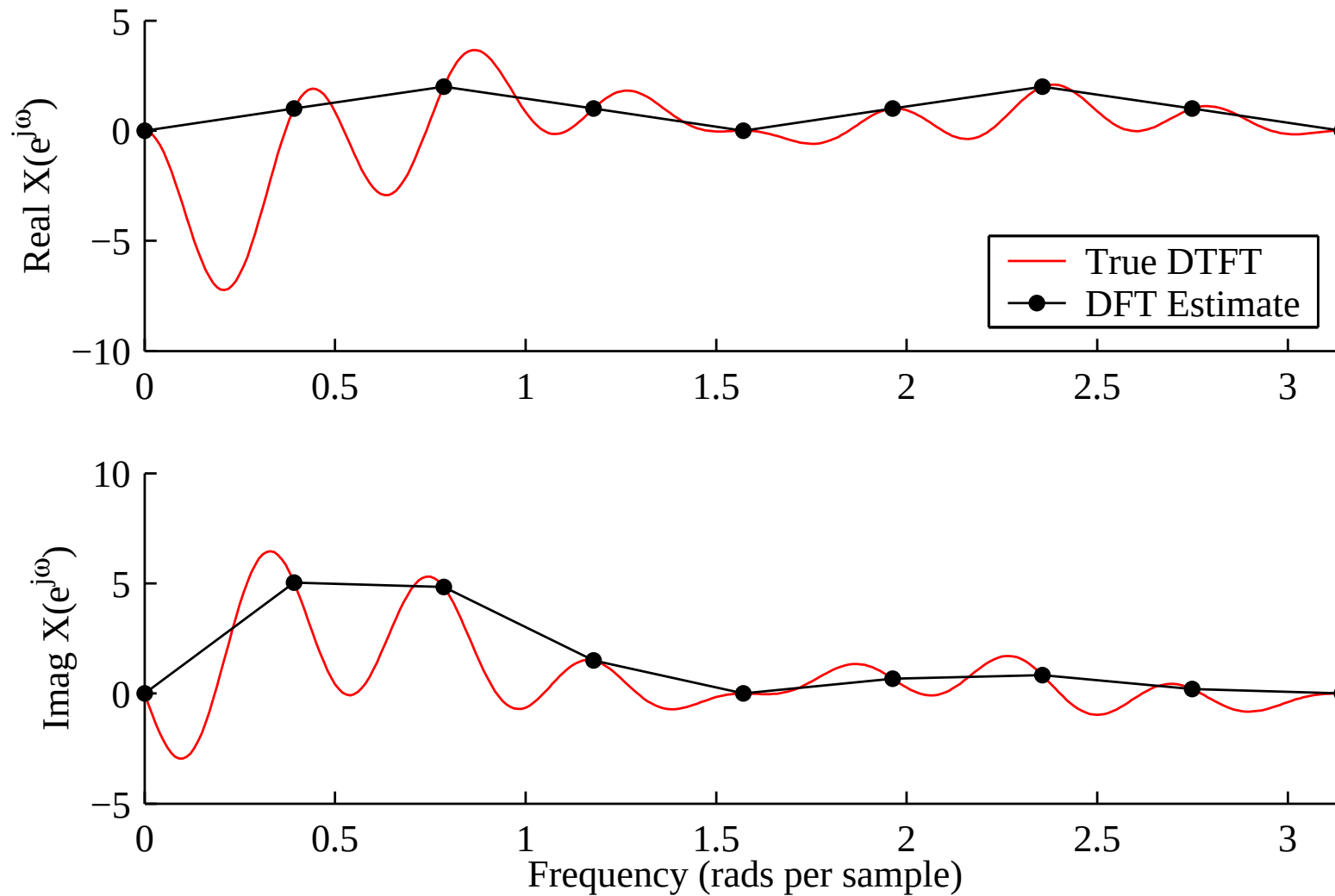
$$X(e^{j\omega}) = \frac{-e^{-j\omega} + e^{-j5\omega} + e^{-j13\omega} - e^{-j17\omega}}{1 - e^{-j\omega}}$$

Use $N = 16$. Repeat with zero padding so that the estimate is evaluated at no less than 1000 frequencies between 0 and π .

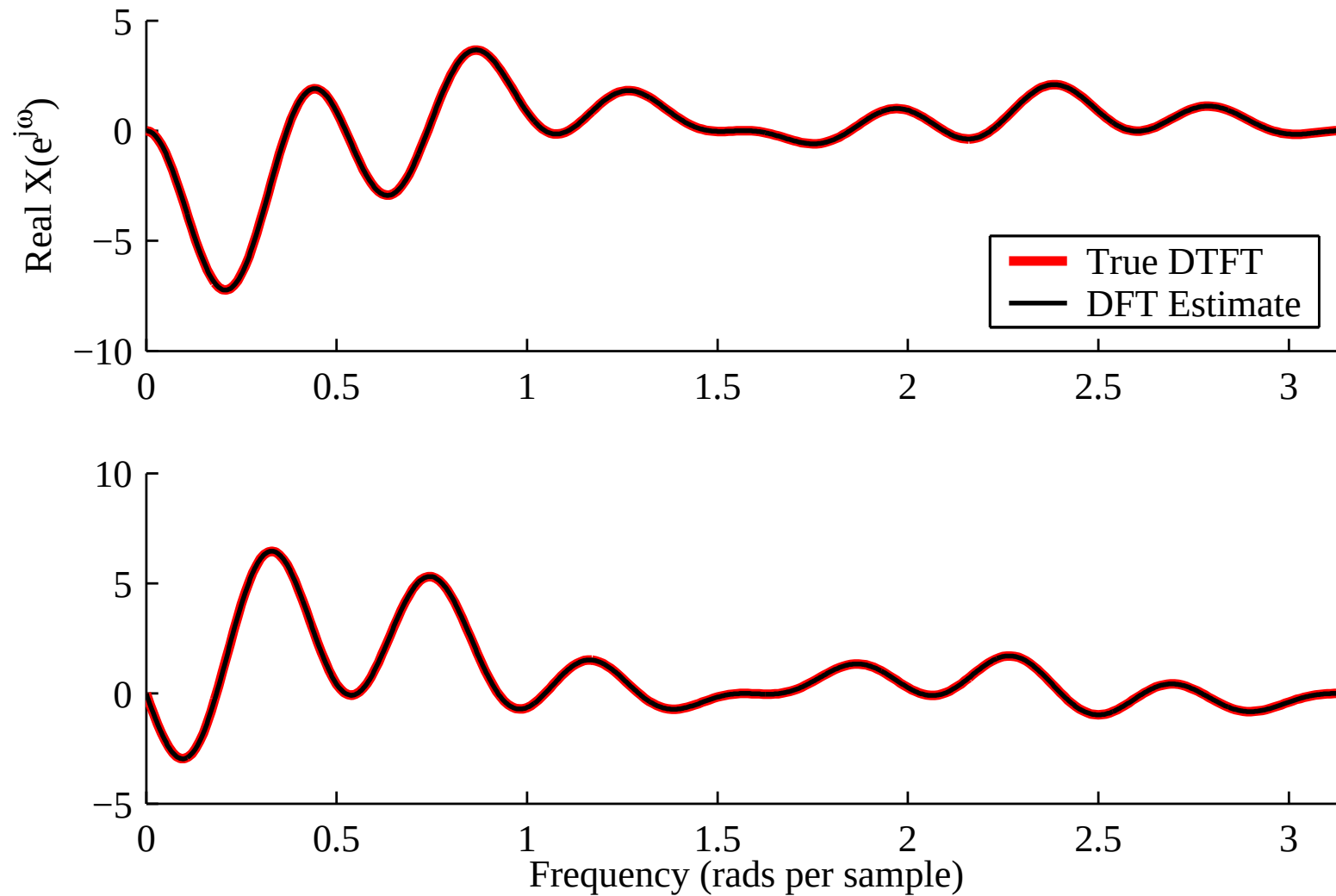
Example 10: Signal



Example 10: DTFT Estimate



Example 10: DTFT Estimate with Padding



Example 10: MATLAB Code

```
function [] = DFTEstimate();
close all;

n = 0:17;
x = -1*(n>=1 & n<=4) + 1*(n>=13 & n<=16);

%=====
% Plot the signal
%=====

figure
FigureSet(1,4.5,2.8);
plot([min(n) max(n)], [0 0], 'k:');
hold on;
    h = stem(n,x, 'b');
    set(h(1), 'MarkerFaceColor', 'b');
    set(h(1), 'MarkerSize', 4);
hold off;
ylabel('x[n]');
xlabel('Time');
xlim([min(n) max(n)]);
ylim([-1.05 1.05]);
box off;
AxisSet(8);
print -depsc FFTESignal;

%=====
% Plot the True Transform \& Estimate
```

```

%=====
n = 1:16;
x = -1*(n>1 & n<=4) + 1*(n>13 & n<=16);
N = length(x);
k = 0:N-1;
we = (0:N-1)*(2*pi/N); % Frequency of estimates
Xe = exp(-j*k*(2*pi/N)*1).*fft(x);
w = 0.0001:0.01:2*pi;
X = (-exp(-j*w*1) + exp(-j*w*5) + exp(-j*w*13) - exp(-j*w*17))./(1-exp(-j*w)); % True spectrum

figure
FigureSet(1,4.5,2.8);
subplot(2,1,1);
h = plot(w,real(X),'r',we,real(Xe),'k');
set(h(2),'Marker','o');
set(h(2),'MarkerFaceColor','k');
set(h(2),'MarkerSize',3);
ylabel('Real X(e^{j\omega})');
xlim([0 pi]);
box off;
AxisSet(8);
legend('True DTFT','DFT Estimate',4);
subplot(2,1,2);
h = plot(w,imag(X),'r',we,imag(Xe),'k');
set(h(2),'Marker','o');
set(h(2),'MarkerFaceColor','k');
set(h(2),'MarkerSize',3); ylabel('Imag X(e^{j\omega})');
xlim([0 pi]);
box off;
xlabel('Frequency (rads per sample)');
AxisSet(8);

```

```

print -depsc FFTEstimate;

%=====
% Plot the True Transform \& Estimate
%=====

n = 1:16;
x = -1*(n>=1 & n<=4) + 1*(n>=13 & n<=16);
N = 2^(ceil(log2(2000)));
k = 0:N-1;
we = (0:N-1)*(2*pi/N); % Frequency of estimates
Xe = exp(-j*k*(2*pi/N)*1).*fft(x,N);

figure
FigureSet(1,4.5,2.8);
subplot(2,1,1);
h = plot(w,real(X),'r',we,real(Xe),'k');
set(h(1),'LineWidth',2.0);
set(h(2),'LineWidth',1.0);
ylabel('Real X(e^{j\omega})');
xlim([0 pi]);
box off;
AxisSet(8);
legend('True DTFT','DFT Estimate',4);
subplot(2,1,2);
h = plot(w,imag(X),'r',we,imag(Xe),'k');
set(h(1),'LineWidth',2.0);
set(h(2),'LineWidth',1.0);
xlim([0 pi]);
box off;
xlabel('Frequency (rads per sample)');
AxisSet(8);

```

```
print -depsc FFTEstimatePadded;
```

z Transform: Definition & Overview

$$\mathcal{Z} \{x(n)\} = X(z) \equiv \sum_{n=-\infty}^{+\infty} x(n) z^{-n}$$

- Discrete-time counter-part of the Laplace transform
- z is complex-valued: $z = re^{j\omega}$
- Primarily analysis of discrete-time LTI systems
- The most important transform is that of the LTI system impulse response $h(n)$
- Trivia: in the field of time series analysis the z -transform is defined as

$$X(z) = \sum_{n=-\infty}^{+\infty} x(n) z^n$$

z Transform: Region of Convergence

$$\mathcal{Z}\{x(n)\} = \sum_{n=-\infty}^{+\infty} x(n) z^{-n}$$

- In general, the z -transform does not converge for all values of z
- The values for which the sum does converge (is finite valued) is called the **region of convergence** (ROC)
- This region is usually shaded on the z -plane
- For example, what is the ROC of
 - $x(n) = \delta(n)$?
 - $x(n) = \delta(n - 1)$?
 - $x(n) = \delta(n + 1)$?
- The ROC (or equivalent) must always be specified along with $X(z)$

z Transform: Region of Convergence Continued

$$\mathcal{Z} \{x(n)\} = \sum_{n=-\infty}^{+\infty} x(n) z^{-n}$$

- In general, the ROC is a ring: $R_1 < |z| < R_2$
- The ROC can give us information about the signal properties and vice versa
 - Finite duration: ROC is entire plane, except possibly $z = 0$ and $|z| = \infty$
 - Causal signals: $0 < R_1 < |z|$ (includes $|z| = \infty$)
 - Anti-causal signals: $|z| < R_2 < \infty$ (includes $z = 0$)
 - Two-sided, infinite duration: possibly $0 < R_1 < |z| < R_2 < \infty$
 - Does it converge for $\cos(\omega_0 n)$? $\cos(\omega_0 n)u(n)$?
 $\cos(\omega_0 n)u(-n)$? Most power signals?
- Book has incorrect extremes for causal and anti-causal signals

z Transform: Relationship to the DTFT

- Can express in terms of the DT Fourier transform

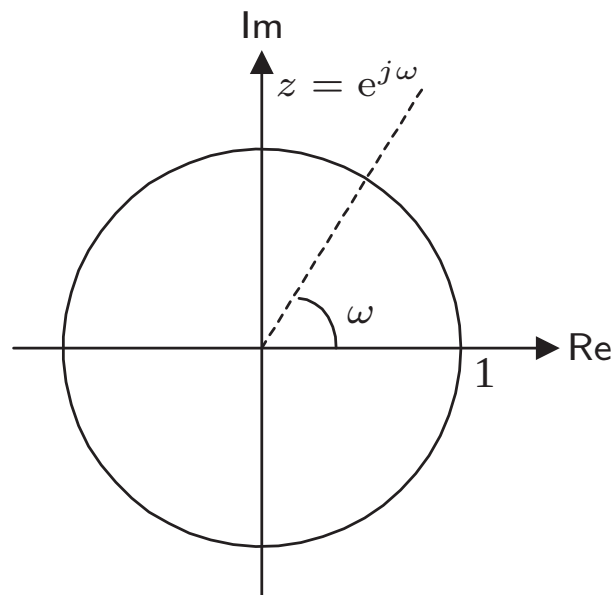
$$\begin{aligned} X(re^{j\omega}) &= \sum_{n=-\infty}^{+\infty} x(n) (re^{j\omega})^{-n} \\ &= \sum_{n=-\infty}^{+\infty} (x(n)r^{-n}) e^{-j\omega n} \\ &= \mathcal{F}\{x(n)r^{-n}\} \end{aligned}$$

- If $|z| = 1$, reduces to the Fourier transform

$$X(z)|_{z=e^{j\omega}} = X(e^{j\omega}) = \mathcal{F}\{x(n)\}$$

- This relationship only holds if the ROC contains the unit circle
 - For example, $u(n) \xleftrightarrow{\mathcal{Z}} U(z)|_{z=e^{j\omega}} \neq \mathcal{F}\{u(n)\}$

z Transform: Complex z Plane



- The z -plane is always drawn with the unit circle $|z| = 1$
- ω is the counter-clockwise angle relative to the positive real axis
- Recall that if $z = e^{j\omega}$, then $X(z)$ is the DTFT
- What value of z corresponds to 0 rad/sample? π rad/sample?

z Transform: Inverse

$$X(z) \triangleq \mathcal{Z}\{x(n)\} = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

$$x(n) \triangleq \mathcal{Z}^{-1}\{X(z)\} = \frac{1}{j2\pi} \oint_C X(z) z^{n-1} dz$$

- The contour of integration is any counterclockwise closed path that encloses the origin and is in the ROC
- The inverse transform is rarely calculated
- We *will not* practice taking transforms or inverse transforms of signals
- We *will* use the z transform to simplify operations such as convolution
- If ROC includes the unit circle $|z| = 1$, then if $z = e^{j\omega}$ we obtain the DTFT analysis and synthesis equations

z Transform: Common Pairs

$$\delta(n) \Leftrightarrow 1 \quad \text{all } z$$

$$u(n) \Leftrightarrow \frac{1}{1 - z^{-1}} \quad |z| > 1$$

$$a^n u(n) \Leftrightarrow \frac{1}{1 - az^{-1}} \quad |z| > a$$

$$r^n \cos(\omega n) u(n) \Leftrightarrow \frac{1 - r \cos(\omega) z^{-1}}{1 - 2r \cos(\omega) z^{-1} + r^2 z^{-2}} \quad |z| > r$$

$$r^n \sin(\omega n) u(n) \Leftrightarrow \frac{r \sin(\omega) z^{-1}}{1 - 2r \cos(\omega) z^{-1} + r^2 z^{-2}} \quad |z| > r$$

Most DSP textbooks have a more complete listing

z Transform: Properties

$$X(z) \triangleq \mathcal{Z}\{x(n)\} = \sum_{n=-\infty}^{\infty} z^{-n}$$

$$x(n) * y(n) \xleftrightarrow{\mathcal{Z}} X(z)Y(z)$$

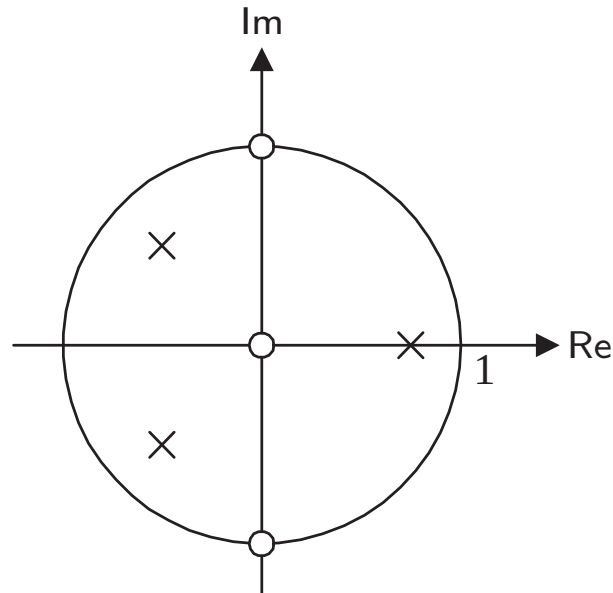
- Properties of the z transform are listed in Table 2.1 (pg. 44) of the text
- The corresponding properties of the DTFT can be obtained by setting $z = e^{j\omega}$
- You should have a working familiarity with z -transforms (or gain one quickly)

z Transform: Rational Functions

$$H(z) = \frac{\sum_{k=0}^Q b_k z^{-k}}{1 + \sum_{k=1}^P a_k z^{-k}} \triangleq \frac{B(z)}{A(z)}$$

- Note that I will use $B(z)$ and b_k for the numerator, rather than $D(z)$ and d_k like the text
- This is a more common notation and consistent with MATLAB's documentation
- If $H(z)$ is a rational function,
 - Roots of $B(z)$ are called **zeros** of $H(z)$
 - Roots of $A(z)$ are called **poles** of $H(z)$
- Note that, by definition, the ROC cannot contain poles
- Technically, $B(z)$ and $A(z)$ must be expressed in terms of positive powers of z to distinguish, for example, a pole at ∞ from a zero at the origin

z Transform: Pole-Zero Plots



- **Pole-zero** plots show where the poles and zeros are on the z -plane
- If the signal is real, poles and zeros always occur in complex conjugate pairs (even symmetry about the real axis)
- Zeros are denoted by $\circ$ and poles are denoted by $\times$
- How do the locations of poles and zeros relate to $|X(e^{j\omega})|$?